

Week 5 Knowledge Check — Technical Infrastructure & Modeling

A short retention check covering the week's core ideas: data models & access patterns, cloud architecture, REST/SOAP APIs, high-level sequence modeling, and performance baselines, monitoring & AI-summary validation. Answer each question, then check yourself against the key at the end. Aim for concepts, not trivia — every answer maps back to something we used in a lab.

Format: 12 questions (multiple choice + true/false). ~15 minutes. Individual or triad.

Questions

1. (MC) In the "Access Pattern First" discipline, what is the *first* step before designing a schema?

- A) Pick the database engine (Postgres by default)
- B) List the queries the feature needs
- C) Draw the entity-relationship diagram
- D) Add indexes to speed up reads

2. (MC) Which storage bucket do you reach for when you need high throughput, simple lookups, caches, and sessions?

- A) Relational (Postgres, MySQL)
- B) Document (MongoDB, Firestore)
- C) Key-value (Redis, DynamoDB)
- D) Search (Elasticsearch, OpenSearch)

3. (T/F) In this course, the TPM's job is to pick the database engine — for example, deciding Postgres vs MongoDB.

4. (MC) In the managed-vs-self "Four-Question Test," which justification for self-managing is called a "smell"?

- A) "The managed service's failure modes don't match our needs"
- B) "Compliance forbids the managed service"
- C) "We want full control"
- D) "Cost is unreasonable at our expected scale"

5. (T/F) "We should be more secure" is a sufficient reason to adopt a customer-VPN-only network boundary.

6. (MC) In the ROM cost table, which cost is called out as the most-forgotten?

- A) Postgres (RDS) hours
- B) Egress bandwidth
- C) S3 storage
- D) Load balancer LCUs

7. (MC) Which HTTP status code signals an idempotency or state collision?

- A) 400 Bad request
- B) 403 Unauthorized
- C) 409 Conflict
- D) 422 Unprocessable

8. (T/F) For a "share" action, `POST /share` (a verb in the URL) is the recommended REST pattern.

9. (MC) In a weird-path sequence diagram, what is described as the "senior-author signal"?

- A) Showing every lifeline in the whole system
- B) Naming the invariant the design preserves
- C) Redrawing the full happy path first
- D) Adding a latency-budget note

10. (T/F) A sad-path sequence that ends with the user stuck (no recovery action) is acceptable as long as the error is rare.

11. (MC) The 4-Question Alert Test says that an alert which can't answer all four questions is:

- A) A page-level (Sev1) alert
- B) Noise — it'll be muted within a sprint
- C) An executive-dashboard metric
- D) A performance baseline

12. (MC) In the AI-Summary Validation Log, which status means "used as input but not yet validated — track to closure"?

- A) Cross-checked all citations
 - B) Spot-checked
 - C) Adopted with rationale
 - D) Pending
-

Answer Key

1. **B** — List the queries first, then design the schema so those queries are cheap. If a data model can't be defended by the queries that drive it, it's wrong.
2. **C** — Key-value (Redis, DynamoDB) is the bucket for high throughput, simple lookups, caches, and sessions. Relational is the B2B default; document is for flexible/hierarchical shapes; search is for full-text.
3. **False** — The TPM doesn't pick the engine. The TPM describes the access pattern clearly enough that the architect can.
4. **C** — "We want full control" is the smell. Default to the managed service; push for the specific decision the managed service forecloses. The other three are legitimate reasons to self-manage.
5. **False** — Don't volunteer for VPN-only; it triples ops complexity. "More secure" is a reason for stronger TLS, not a customer-VPN boundary — only choose it if a customer requires it.
6. **B** — Egress bandwidth is the most-forgotten cost. The ROM goal is rough order of magnitude; the TPM job is catching the 2x surprise before it ships.
7. **C** — 409 Conflict is idempotency / state collision. (401 = unauthenticated; 403 = authenticated but lacks permission; 422 = validated but violates a business rule.)
8. **False** — URLs are nouns; the verb is the HTTP method. `POST /share` is wrong; `POST /shares` is right.
9. **B** — Naming the invariant (e.g., "at most one ReconcileEvent per Idempotency-Key per 24h") is the senior-author signal. If a triad can't name the invariant, the weird-path design isn't done.
10. **False** — The user must have a path forward, not a dead-end. A sad path that ends "user is stuck" is not designed — force the recovery action.
11. **B** — If you can't answer all four (meaning, on-call action, threshold + window, severity), the alert is noise and will be muted within a sprint.
12. **D** — "Pending" means used as input but not yet validated; it needs a deadline and must be tracked to closure, or it becomes "we trusted the AI."